

4.4 Taylor Theorem Worksheet

1. Use an appropriate Taylor polynomial of degree 2 to obtain a good estimate for $\sqrt{403}$.

We will construct a Taylor polynomial centered at 400 to estimate this value.

So we want to find:
$$p_2 = \frac{f(400)}{0!} + \frac{f'(400)}{1!}(x-400) + \frac{f''(400)}{2!}(x-400)^2$$

We note that:

$$f(x) = \sqrt{x} \rightarrow f(400) = 20$$

$$f'(x) = \frac{1}{2}x^{-1/2} \rightarrow f'(400) = \frac{1}{40}$$

$$f''(x) = -\frac{1}{4}x^{-3/2} \rightarrow f''(400) = \frac{-1}{32000}$$

Therefore we have:

$$f(x) \approx p_2 \text{ which mean } \sqrt{x} \approx 20 + \frac{1}{40}(x-400) + \frac{-1/32000}{2!}(x-400)^2 \text{ for values of } x \text{ near } 400.$$

$$\text{Thus } f(403) = \sqrt{403} \approx 20 + \frac{1}{40}(3) - \frac{1}{64000}(3)^2 = 20 + \frac{3}{40} - \frac{9}{64000}$$

2. Consider the function $f(x) = e^{-x}$.

- a) Use an appropriate Taylor polynomial of degree 2 to obtain a good estimate for $e^{-1/2}$.

We will construct a Taylor polynomial centered at 0 to estimate this value.

So we want to find:
$$p_2 = \frac{f(0)}{0!} + \frac{f'(0)}{1!}(x-0) + \frac{f''(0)}{2!}(x-0)^2$$

We note that:

$$f(x) = e^{-x} \rightarrow f(0) = 1$$

$$f'(x) = -e^{-x} \rightarrow f'(0) = -1$$

$$f''(x) = e^{-x} \rightarrow f''(0) = 1$$

Therefore we have:

$$f(x) \approx p_2 \text{ which mean } e^{-x} \approx 1 - (x-0) + \frac{1}{2!}(x-0)^2 = 1 - x + \frac{1}{2}x^2 \text{ for values of } x \text{ near } 0.$$

$$\text{Thus } f(0.5) = e^{-0.5} \approx 1 - (0.5) + \frac{1}{2}(0.5)^2 = \frac{1}{2} + \frac{1}{8} = \frac{5}{8} = 0.625$$

b) Use Taylor's Theorem to find a bound on the error of this approximation.

We note that $f(x) = e^{-x}$ is infinitely differentiable at all points and that on the interval $[0, 0.5]$, $|f^{(n+1)}(x)| \leq e^0 = 1 = M$. Therefore, we have satisfied the conditions to use Taylor's Inequality.

Taylor's Inequality states that $|R_n(x)| \leq \frac{M}{(n+1)!} |x - x_0|^{n+1}$. Note that we have the following values based on our work in part (a): $x = 0.5$, $x_0 = 0$, & $n = 2$. Therefore we have,

$$|R_2(0.5)| \leq \frac{1}{(2+1)!} |0.5 - 0|^{2+1}$$

$$\Leftrightarrow |R_2(0.5)| \leq \frac{1}{6} \cdot \frac{1}{8} = \frac{1}{48}$$

Therefore we know that the error of the approximation $e^{-0.5} \approx 0.625$ is no larger than $\frac{1}{48}$.

c) Use Taylor's Theorem to show that $f(x) = e^{-x}$ is equal to its Taylor's series, centered at x_0 , for all x .

We note that $f(x) = e^{-x}$ is infinitely differentiable at all points and that for some real number x there exists a number d such that $|x| \leq d$. Then clearly we have that, $|f^{(n+1)}(x)| \leq e^{-(-d)} = e^d = M$.

Now by Taylor's Inequality we have $0 \leq |R_n(x)| \leq \frac{e^d}{(n+1)!} |x - x_0|^{n+1}$ for $|x| \leq d$ and each value of n .

We note that $\lim_{n \rightarrow \infty} \frac{e^d}{(n+1)!} |x - x_0|^{n+1} = e^d \lim_{n \rightarrow \infty} \frac{|x - x_0|^{n+1}}{(n+1)!} = e^d \cdot 0 = 0$ (again using 3.1.2 example 5).

Therefore, by the Squeeze Theorem for sequences we have that $\lim_{n \rightarrow \infty} |R_n(x)| = 0$ for $|x| \leq d$ and each value of n which immediately implies that $\lim_{n \rightarrow \infty} R_n(x) = 0$ for $|x| \leq d$ and each value of n .

Thus, $f(x) = e^{-x}$ is equal to its Taylor series, centered at x_0 , for all x .

d) How many terms of the Taylor series centered at 0 are needed to estimate $e^{-1/2}$ accurately to 3 decimal places?

We know that $e^{-x} = \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}$. We want to choose n so that the absolute value of the n th remainder at

$x = \frac{1}{2}$ satisfies: $|R_n(0.5)| < 0.0005$. By Taylor's Inequality we know that

$$|R_n(0.5)| \leq \frac{1}{(n+1)!} |0.5 - 0|^{n+1} = \frac{1}{(n+1)!} \frac{1}{2^{n+1}} < \frac{1}{(n+1)!}. \text{ Therefore we want:}$$

$$\frac{1}{(n+1)!} < 0.0005 \Leftrightarrow 2000 < (n+1)! \Leftrightarrow n \geq 6$$

So obtain an approximation of $e^{-1/2}$ that is accurate to 3 decimal places we must use at least the first 7 terms of the sequence (remember that the sequence here starts with $n = 0$).

3. The function $f(x) = \cosh(x)$ has a Taylor series about 0 that is $\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$.
- a) Use the first 3 terms of this series to obtain an approximation for $\cosh(1)$.

$$\cosh(1) \approx 1 + \frac{1}{2} + \frac{1}{24} = \frac{37}{24}$$

- b) Use Taylor's Inequality to determine an upper bound for the error of this approximation.

We note that $f(x) = \cosh(x)$ is infinitely differentiable at all points and that $f^{(n+1)}(x)$ is either $\cosh(x) = \frac{e^x + e^{-x}}{2}$ or $\sinh(x) = \frac{e^x - e^{-x}}{2}$. If we look on the interval $[0, 1]$ then clearly,

$$|f^{(n+1)}(x)| \leq \cosh(1) = \frac{e + e^{-1}}{2} = M.$$

Therefore, we have satisfied the conditions to use Taylor's Inequality.

Taylor's Inequality states that $|R_n(x)| \leq \frac{M}{(n+1)!} |x - x_0|^{n+1}$. Note that we have the following values based on our work in part (a): $x = 1$, $x_0 = 0$, & $n = 4$. Therefore we have,

$$|R_4(1)| \leq \frac{\left(\frac{e + e^{-1}}{2}\right)}{(4+1)!} |1 - 0|^{4+1} = \frac{e + e^{-1}}{240} < \frac{3 + 1}{240} = \frac{1}{60}$$

Therefore we know that the error of the approximation $\cosh(1) = \frac{37}{24}$ is no larger than $\frac{1}{60}$.

- c) How many terms of the Taylor series are needed to estimate $\cosh(1)$ accurately to 5 decimal places?

We know that $\cosh(x) = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$. We want to choose n so that the absolute value of the n th remainder at $x = 1$ satisfies: $|R_n(1)| < 0.000005$. Note that any derivative of $\cosh(x)$ is either $\cosh(x)$ or $\sinh(x)$ and we can see that $\cosh(x) \geq \sinh(x)$ because $\frac{e^x + e^{-x}}{2} \geq \frac{e^x - e^{-x}}{2}$. So, by Taylor's Inequality we

know that $|R_n(1)| \leq \frac{\cosh(1)}{(n+1)!} |1 - 0|^{n+1} = \frac{\left(\frac{e^1 + e^{-1}}{2}\right)}{(n+1)!} < \frac{2}{(n+1)!}$. Therefore we want:

$$\frac{2}{(n+1)!} < 0.000001 \Leftrightarrow 2,000,000 < (n+1)! \Leftrightarrow n \geq 9$$

So obtain an approximation of $\cosh(1)$ that is accurate to 5 decimal places we must use at least the first 10 terms of the sequence (remember that the sequence here starts with $n = 0$).